Dynamic voltage can also be reduced for supply voltage, which again has its downside. Reduction in supply voltage increases the delay of the cell as well. A balance has to be maintained between supply voltage and device performance.

Static voltage scaling presents a few constraints while operating electronics. It does not allow changes in the supply blocks once designed, hence, is not adaptive.

Adiabatic circuits are interesting in this regard. Here, energy spent to flip a bit can be reduced to very small values by externally controlling the length and shape of signal transitions.

An IC to reduce power consumption by 90 per cent. An adiabatic logic circuit can use AC voltage as its power source without converting it to DC voltage. Yasuhiro Takahashi, department of electrical electronic and computer engineering, Gifu University, Japan, has built a prototype that could be installed in healthcare products. The 155µm × 172µm chip operates on AC.

The need arose from central processing units (CPUs) operating at high frequency levels of 500MHz, thereby rendering the circuits not advantageous for computing purposes. Interestingly, such a circuit could be used in healthcare devices or wristwatches that did not process at high speed. However, there are other methods for regular applications.

Low-power design techniques

Transistors are usually arranged in a plane structure, but applying a spatial structure and the adiabatic logic circuit could reduce power loss and thus power consumption. "Powering down various modes, not adding unnecessary parts and others must be taken care of while designing the circuit," explains Anand. He adds, "Besides the hardware design and supporting components, overall functioning and the firmware code also contribute to power saving of the device."

Either through architecture design that uses low-power techniques, or by adopting a process that can reduce consumption, designers always look for ways to reduce power consumption. Kamat explains, wireless systems consume a lot of power in transmission. He says, "The rate of transfer of information can be controlled and helps in minimising power consumption. Sampling rate of power consumption is reduced to increase power capacity." Some of the solutions come at an expense in performance, reliability or chip area.

Making compromises in system design. Changing system architecture has been the most common technique for reducing power consumption. Clock gating is a very popular dynamic power-reduction technique. Power consumed during device logic switching and charging load capacitance are the major areas of concern.

Latch based clock gating is an interesting technique that saves designs from hazards that can lead to additional power consumption. The clock can be turned off when not needed. Modern EDA tools identify the circuits where clock gating can be inserted.

Traditional methods control the selection on a multiplexer. Latch based clock gating adds a level-sensitive latch to circuit design. It enables signal from the active edge of the clock until the inactive edge of the clock. Interestingly, clock gating does not require modifications to the RTL description.

Selection of components. "Selection of components for low-power devices involves two critical parameters," explains Anand. Low quiescent current as well as lower operating power need to be calculated, and the component's power should be controllable. "Controlling the component allows turning off the device, put it into lower-power mode or disable it using external control."

Multiple voltages. Another trick to save power is to operate different areas of the circuit at different voltages. In any circuit, there are specific areas that operate at high voltages, whereas other parts can work well at lower voltages for major power savings. This also reduces leakage power.

Power gating is a technique implemented in devices with sleep mode. In certain instances, only part of the device is required to function. It makes sense to switch off non-functional blocks to save power. Leakage power and dynamic consumption can be reduced in such power-gated blocks. However, power coming from power-reduced blocks should not affect the functioning blocks. Hence, the design has to incorporate isolation blocks, so that functionality corruption does not occur.

The demand

Mobile device consumers require longer talk and standby times with extended battery life at lower cost. Demand for smaller and sleeker devices has been resulting in higher levels of silicon integration in advanced processes. However, higher leakage current in advanced processes requires a reduction to reduce power consumption. Losses through heat are also being managed with some ingenious solutions. "Energy dissipation in the form of heat can also be used to generate power," says Kamat. EFY

Major sources of power dissipation

Power dissipation in electronics can be classified into three major categories:

Leakage current. Diode leakages, sub-threshold leakage, gate leakage and tunnel currents are the major areas of current leakages that occur even in standby mode.

Dynamic power consumption. Logic transitions cause gates to charge/discharge load capacitance.

Short-circuits. Logic gate state changes in CMOS result in momentary short-circuits, leading to power dissipation to the extent of ten per cent of the total power.